# RANJIT PATIL

Full Stack Developer

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## PROFESSIONAL SUMMARY

Final-year Computer Science and Engineering student graduating in 2027, with hands-on experience building web applications using React, TypeScript, Java Spring Boot, and Python FastAPI. Have worked on REST APIs, user authentication, database integration, and real-time application features. Also familiar with Docker and Redis, with an interest in developing reliable and user-friendly software.

## EDUCATION

**Bachelor of Engineering – Computer Science and Engineering (Honors - AI & ML)** **2023–2027 | CGPA: 7.52**
*SKN Sinhgad College of Engineering, Pandharpur*

**Higher Secondary Certificate (HSC – 12th)** **Percentage: 75.67%**
*Ligade Patil Junior College of Science, Karad*

**Secondary School Certificate (SSC – 10th)** **Percentage: 81.40%**
*Shivanjali English Medium School, Navarasta*

## TECHNICAL SKILLS

**Languages:** Java, JavaScript (ES6+), TypeScript, Python, SQL, HTML5, CSS3

**Frontend:** React.js, Next.js, React Query, Redux Toolkit, Tailwind CSS, shadcn/ui, Vite, MapLibre GL JS, Recharts

**Backend:** Spring Boot, Node.js, Express.js, FastAPI, REST APIs, Socket.io, JPA/Hibernate, SQLAlchemy, JWT, Role-Based Access Control (RBAC)

**Databases & Messaging:** PostgreSQL, PostGIS, MySQL, MongoDB, Neo4j, Redis, Redpanda (Kafka)

**DevOps, Testing & Tools:** Docker, Docker Compose, Nginx, Flyway, Alembic, Git, GitHub Actions, Pytest, Vitest, JUnit 5, Linux

**Core CS:** Data Structures & Algorithms, Object-Oriented Programming (OOP), DBMS, Operating Systems, Computer Networks, Concurrency Control

## INTERNSHIP EXPERIENCE

**Full Stack Development Intern | Cognifyz IT Solutions Pvt. Ltd. | Remote** **Oct 2026 – Present**

- Develop responsive frontend components using React and TypeScript and integrate them with backend REST APIs.
- Implement form validation, API data fetching, and authentication-related features while following the application's access-control requirements.
- Work with relational databases to design schemas, write SQL queries, and troubleshoot API and database integration issues.

## PROJECTS

**TicketReserve | Event Ticketing & Seat Reservation Platform [GitHub]** **2026**
*[ React, TypeScript, Tailwind CSS, React Query, Java 17, Spring Boot, Spring Security, JWT, JPA/Hibernate, MySQL, Redis, Flyway, Docker ]*

- Built a full-stack ticket booking application with seat selection, booking management, and checkout features using React and Spring Boot.
- Prevented duplicate seat reservations through Redis atomic locks, JPA optimistic locking, and MySQL constraints.
- Implemented booking status transitions, idempotency checks for repeated requests, and automatic release of expired seat holds.
- Secured APIs using JWT authentication, role-based access control, and booking ownership checks.

**Aegis Cloud Telemetry | IoT Fleet Management Platform [GitHub]** **2026**
*[ React, TypeScript, MapLibre GL JS, Turf.js, Node.js, Express, Socket.io, Redpanda (Kafka), Redis, MongoDB, Docker ]*

- Built a real-time fleet monitoring platform that simulates telemetry from 1,000 trucks per second and displays vehicle locations on an interactive map.
- Implemented live location updates and geofence alerts using Socket.io, Redis Pub/Sub, MapLibre GL JS, and Turf.js.
- Added historical route playback by retrieving vehicle location records from MongoDB and displaying routes on the map.
- Implemented Kafka-based error handling with a dead-letter queue and retry worker, along with temperature-trend monitoring to flag potential engine overheating.

**EpiGuard AI | Community Health Monitoring & Early-Warning Platform [GitHub]** **2026**
*[ React, TypeScript, Tailwind CSS, FastAPI, Python, SQLAlchemy, PostgreSQL, PostGIS, Redis, XGBoost, Docker ]*

- Developed a public-health decision-support platform to help health officials monitor water-quality data and identify areas at risk of water-borne diseases.
- Built role-based dashboards for administrators, health officials, analysts, and community workers, with regional risk visualization using PostGIS.
- Integrated data validation, anomaly detection, and machine-learning workflows to support risk analysis across live, demo, and simulation modes.
- Co-authored a published research paper and registered a patent based on the platform's predictive community health monitoring workflow.

## CERTIFICATIONS, PUBLICATIONS & ACHIEVEMENTS

**Patent & Publication:** Published research paper and registered patent for *"Smart Community Health Monitoring and Early Warning Platform"* (EpiGuard AI, 2026).

**Neo4j Certified Professional ↗:** Graph databases and Cypher, GraphAcademy (Oct 2026).

**Agentic AI Certified Foundations Associate ↗:** Oracle University (Oct 2026).

**NPTEL Cloud Computing (Elite Certificate) ↗:** IIT Kharagpur (2026).

**TATA Group Virtual Job Simulations (Forage):** GenAI Powered Data Analytics ↗ | Cybersecurity IAM Strategy ↗ (Oct 2026).

## LANGUAGES

**English:** Full professional proficiency | **Hindi:** Full professional proficiency | **Marathi:** Native